

# Shoulder Anatomy and Biomechanics

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The shoulder can really be considered a complex (Fig. 36.1), consisting of four joints or articulations (all with important stabilizing ligaments), two spaces, and more than 30 muscles and their respective tendons. The shoulder complex is an intricate structure that requires synchronized/orchestral-type motions/movements to function properly. A thorough understanding of the anatomy and complex biomechanics of the shoulder is helpful to clinicians in diagnosing disorders, applying appropriate surgical procedures, and implementing proper rehabilitation protocols.

## PERTINENT BONY ANATOMY

The proximal humerus consists of the humeral head and articular surface, the greater and lesser tuberosities, which are the attachment sites for the rotator cuff muscles, and the humeral shaft. Between the tuberosities is the bicipital groove, the location of the tendon of the long head of the biceps brachii muscle, as it exits the glenohumeral joint. The articular cartilage on the humeral head has been shown to vary from 0.2 to 2.0 mm and is thickest in the central portion of the humeral head.<sup>1</sup> Multiple anatomic and radiologic studies have been performed in an attempt to better define the relationships and geometry of the various parts of the proximal humerus, glenoid, and glenohumeral joint.<sup>2–8</sup> Different measuring techniques and reference points make comparisons between the studies difficult. The inclination of the humeral head articular surface, as referenced to the humeral canal, varies from 30 to 55 degrees.<sup>3,6–8</sup> The mean radius of curvature of the humeral head is  $24 \pm 1.2$  mm but can be as much as 30 mm.<sup>4</sup> The superior-most portion of the humeral head is a mean of 8 mm higher than the greater tuberosity. This relationship varies between men and woman but is proportional to the radius of curvature of the articular surface.<sup>4</sup> Retroversion of the humeral head, in reference to the center of the humeral canal, has been shown to be quite variable, ranging from 0 to 55 degrees (Fig. 36.2).<sup>6,7,9</sup> Medial and posterior offset of the humeral head is variable with reference to the humeral shaft.<sup>3,6–8,10</sup>

The scapula is a relatively flat, triangular-shaped bone that is positioned on the posterolateral aspect of the thorax at the level of the second and seventh ribs. The scapula has three angles—superior, inferior, and lateral. The lateral angle, formed by the superior and lateral borders of the scapula, gives rise to the glenoid. Anatomic variants at the superior angle are sometimes the cause of snapping scapula syndrome.<sup>11</sup> The scapular spine divides the dorsal aspect into the supraspinatus and

infraspinatus fossa. The subscapularis fossa is located on the ventral or anterior portion of the scapular body. Several muscles originate on the scapula, namely the supraspinatus, infraspinatus, subscapularis, teres major and minor, triceps, and deltoid muscles. Several other important muscles insert on the scapula: the serratus anterior, levator scapulae, rhomboid major and minor, trapezius, pectoralis minor, short head of the biceps brachii, and coracobrachialis muscles.

The scapula has two processes or projections: the coracoid and the acromion. The coracoid projects from the superior and lateral aspect and projects anteriorly and laterally. The mean distance from the coracoid tip to the anterior border of the coracoclavicular ligament can vary from 22 to 28 mm.<sup>12,13</sup> The mean width of the coracoid is 15.9 mm, and the mean thickness is 10.4 mm.<sup>13</sup> The use of the coracoid as an extension of the glenoid for shoulder stabilization procedures is well described<sup>14,15</sup> and has, once again, become a treatment option for patients with glenohumeral instability and bone deficiencies. The acromion process is the lateral and anterior extension of the scapular spine. The anteroinferior aspect of the acromion serves as the attachment site for the coracoacromial ligament (Fig. 36.3). The morphology of the acromion has been studied, and some correlations have been made between aggressive anterior acromial hooks/spurs and rotator cuff tears.<sup>16–20</sup> Some evidence indicates that the coracoacromial ligament helps provide superior stability to the glenohumeral joint.<sup>21,22</sup> The lateral portion of the coracoacromial ligament has been found to have decreased mechanical properties, is shorter in length, and has a larger cross-sectional area in patients with rotator cuff tears.<sup>19</sup> Resection of the coracoacromial ligament can lead to anterosuperior humeral head escape in patients with massive rotator cuff tears. Biomechanical studies have shown that the rotator cuff becomes closest to the undersurface of the acromion between 60 and 120 degrees.<sup>23</sup> The subacromial bursa is located in the anterior portion of the subacromial space and is under the coracoacromial arch and deltoid. The bursa helps reduce friction between the coracoacromial arch and the rotator cuff when the arm is elevated but can be impinged under the acromion in certain conditions. The bursa has significant nerve endings, including Ruffini endings, Pacinian corpuscles, and free nerve endings,<sup>24,25</sup> and is a source of pain in the subacromial space.

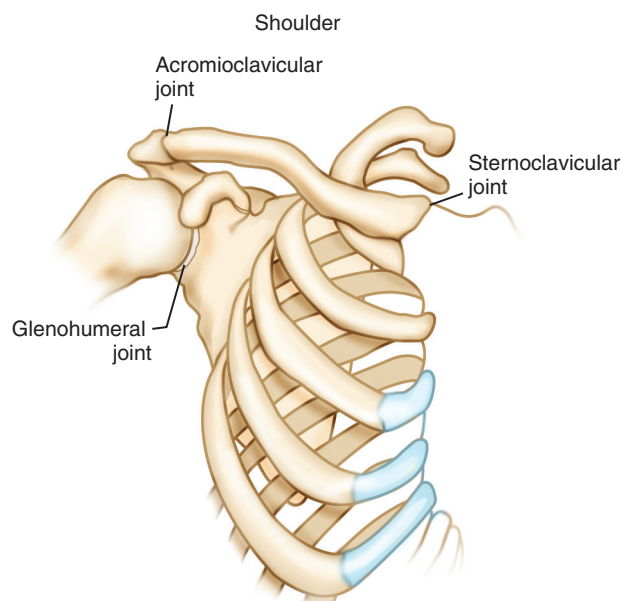
Just medial to the coracoid and anterior to the supraspinatus fossa is the suprascapular notch of the scapula. As the suprascapular nerve travels from the upper trunk of the brachial plexus,

**Abstract**

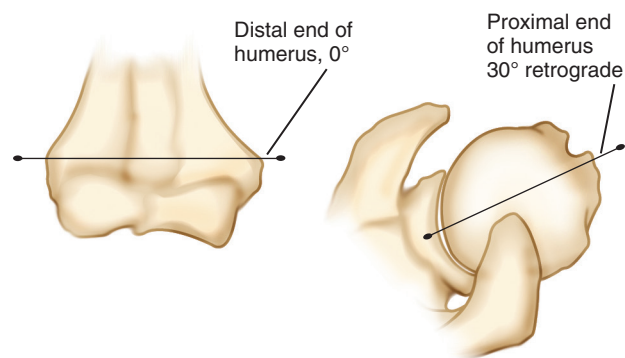
The shoulder is a complex arrangement of four bones, four articulations, and several muscles, tendons, and ligaments. The shoulder complex requires healthy and ligamentously stable joints and numerous muscles working in synergy to allow the upper extremity to be placed in various positions and to perform activities that subject it to extremes of force and torque. The various articulations and muscles are kinetically linked, and abnormalities in any of the joints or muscles can result in pain, dysfunction, and possibly injury to other structures. An appreciation of the anatomy and biomechanics of the shoulder complex will be beneficial to clinicians in making diagnoses, applying safe and appropriate surgical procedures, and implementing appropriate rehabilitation protocols.

**Keywords**

humerus  
scapula  
glenoid  
rotator cuff  
glenohumeral joint



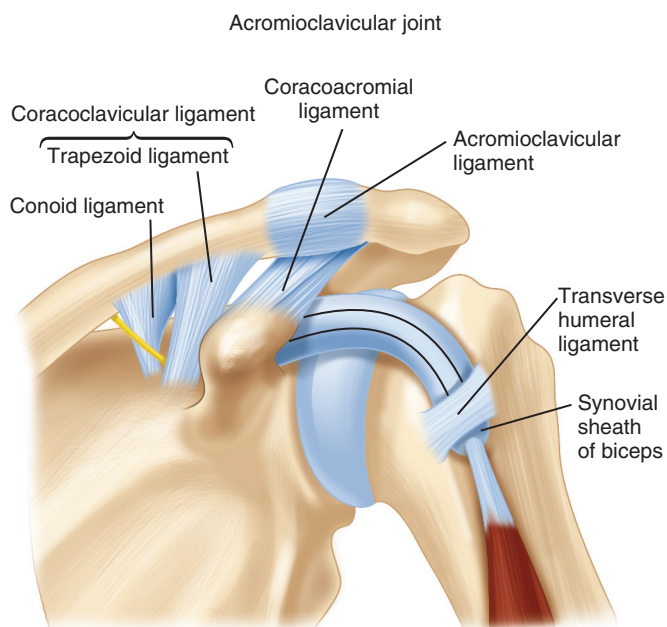
**Fig. 36.1** The shoulder complex consists of the glenohumeral, acromioclavicular, and sternoclavicular joints, along with the scapulothoracic articulation.



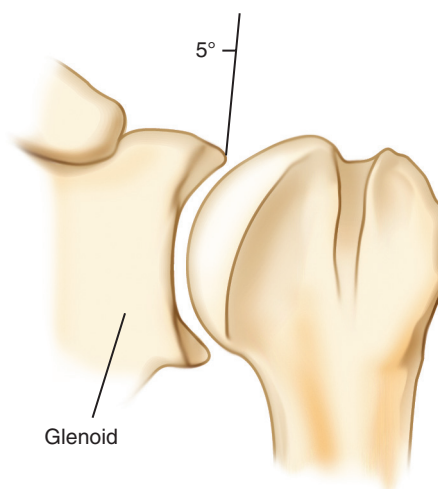
**Fig. 36.2** With respect to the epicondylar axis of the distal humerus, the humeral head articular surface is retroverted from 0 to 55 degrees, averaging 30 degrees.

it enters this notch prior to innervating the supraspinatus. The nerve branches to innervate the supraspinatus within 1 cm of the notch.<sup>26</sup> The superior transverse scapular ligament is the roof of the suprascapular notch. Anatomic studies have shown that the notch is U-shaped in three-fourths of specimens. The superior transverse scapular ligament can be partially or completely ossified in some people.<sup>27</sup>

The lateral angle of the scapula is the location of the glenoid. The glenoid has an average of 5 degrees of superior tilt, referencing the scapula (Fig. 36.4).<sup>28</sup> The superior slope to the glenoid plays a role in preventing inferior subluxation of the humerus. The glenoid is also retroverted with respect to the transverse axis of the scapula. Measurements vary depending on the imaging study used to perform the measurements. However, most studies have shown that the glenoid has 1 to 3 degrees of retroversion<sup>29,30</sup>; however, retroversion can vary from 14 degrees of anteversion to 12 degrees of retroversion in normal shoulders.<sup>31</sup> Retroversion may be overestimated on plain axillary radiographs and



**Fig. 36.3** An anterior view of the shoulder demonstrating the coracoclavicular ligaments, acromioclavicular joint, coracoacromial ligament, and transverse humeral ligament.



**Fig. 36.4** Relative to the plane of the scapula, the fossa is angled slightly inferior and posterior, offering little bony support to inferior instability with the arm at the side.

is probably more accurately measured by computed tomography.<sup>30</sup> The scapula is anteverted approximately 30 degrees to the coronal plane. The glenoid is shaped like a pear; it is wider inferiorly. The average superior-inferior length of the glenoid is 39 mm, and the anteroposterior width in the lower half averages 29 mm.<sup>4,32</sup> The glenoid has a bare spot, which has been shown to be in the center of the lower portion on the glenoid. The bare spot can be used as a reference in measuring anteroinferior glenoid bone loss.<sup>33</sup> The radius of curvature of the glenoid is, on average, 2.3 mm more than the humeral head.<sup>4</sup> Unlike the humeral head, where the articular cartilage is thickest in the center, the articular cartilage is thickest on the periphery of the glenoid and thinnest in the center.

The spinoglenoid notch is an indentation at the confluence of the lateral edge of the base of the scapular spine and glenoid neck. This notch connects the supraspinatus and infraspinatus fossae. The suprascapular nerve and vessels travel through this notch prior to the nerve innervating the infraspinatus. On average, the suprascapular nerve is 1.5 cm medial to the posterior glenoid rim.<sup>26</sup> Knowledge of the location of this nerve is important when one is directly dissecting in the posterior aspect of the shoulder and when placing screws from anterior to posterior across the glenoid.<sup>34</sup> Compression of the suprascapular nerve in this location can result in isolated weakness of the infraspinatus muscle. The spinoglenoid notch can have a ligament across it. This ligament is present in 14% to 61% of people.<sup>27,35</sup>

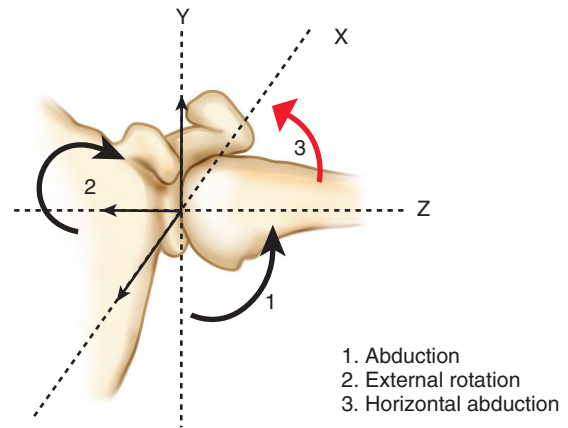
The clavicle is an S-shaped or double-curved bone that connects the shoulder complex to the axial skeleton via the sternoclavicular joint. It is formed by membranous bone but does have a physis at the medial aspect. As a strut from the sternum to the shoulder, it is important in maintaining proper scapular positioning and kinematics during shoulder movement. Shortening of the clavicle, as is seen in some clavicular fracture malunions, leads to significant changes in shoulder posture and scapular positioning during shoulder motions.<sup>36,37</sup> In addition to its role in shoulder kinematics, the clavicle protects vascular structures to the upper extremities and the brachial plexus. The clavicle is the site of origin of a portion of the deltoid and pectoralis major muscles, as well as the insertion site of a portion of the trapezius muscle; all these muscles influence shoulder motion.

## JOINTS AND ARTICULATIONS OF THE SHOULDER COMPLEX

### Glenohumeral Joint

The glenohumeral joint is the most mobile joint in the body. The joint has 6 degrees of freedom, allowing for glenohumeral joint translations and rotations. Shoulder joint rotations occur in the coronal plane and are commonly referred to as abduction and adduction. Rotations in the sagittal plane are called flexion and extension. Rotations relative to the long axis of the humerus are called internal and external rotation (Fig. 36.5). The normal shoulder has substantial translational motion, with as much as 8 to 14 mm of translation anterior, posterior, and inferior with manual clinical testing.<sup>38,39</sup> Examinations with use of an anesthetic have documented even greater laxity, with subluxation over the glenoid anteriorly in 81.6% of subjects and posteriorly in 57.5% of subjects.<sup>40</sup> Given the mobility and wide range of motion, it is really not surprising that the shoulder is also the most unstable joint in the body. Shoulder stability is achieved through a combination of inherent joint characteristics, static stabilizers, and muscular or dynamic stabilizers.

The shoulder is afforded some stability from the inherent negative intraarticular pressure. This negative articular pressure is due to the glenoid concavity's "plunger" effect on the humeral head. The loss of the intra-articular vacuum as a result of an opening in the joint capsule results in less shoulder stability.<sup>41,42</sup> Some inherent stability is also achieved as a result of an adhesion-cohesion effect that occurs when two wet surfaces come in contact. Concavity-compression is another mechanism that has been



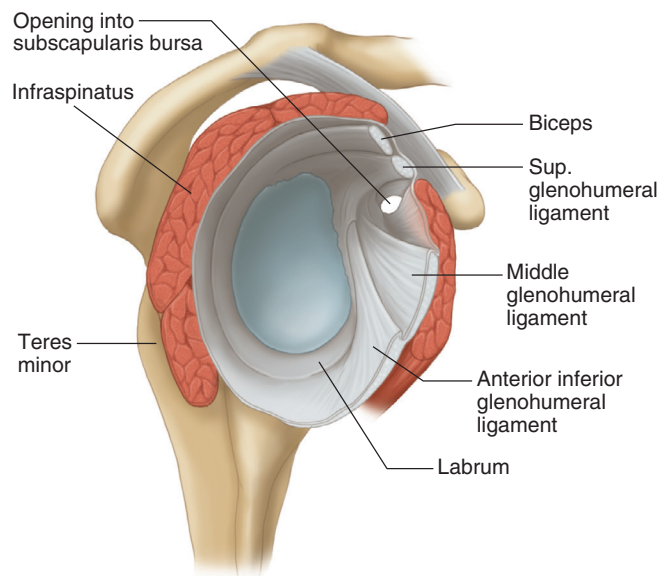
**Fig. 36.5** Shoulder motions are rotations along an axis. Rotations along the X axis or coronal plane are referred to as abduction and adduction. Rotations along the long axis of the humerus, when the arm is at the side, occur through the Y axis and are referred to as internal and external rotation. Rotations in the sagittal plane or Z axis (when the arm is at the side) are referred to as flexion and extension.

found to play a role in glenohumeral stability.<sup>43</sup> A compressive load provided by the rotator cuff forces the convex humeral head into the concave glenoid.

Static stability of the glenohumeral joint is provided by the glenoid labrum, glenohumeral ligaments, and joint capsule. The glenoid labrum is a fibrocartilaginous structure that is attached along the periphery of the glenoid. The labrum is wedge shaped, which increases the effective depth of the glenoid by approximately 50%.<sup>44</sup> Increasing the concavity of the glenoid contributes to overall shoulder stability because of the concavity-compression and the suction effect as a result of the intraarticular vacuum that occurs. The labrum contributes to glenohumeral stability by providing a bumper effect to prevent abnormal translations of the humeral head. The morphology and attachment of the labrum varies in the different quadrants of the shoulder. Inferiorly, the labrum is well attached to the glenoid, becoming an extension of the articular cartilage.<sup>45</sup> The firm attachment in the anteroinferior quadrant increases the diameter of the glenoid and increases the contact surface area. Removal of the antero-inferior labrum results in up to a 15% loss in glenohumeral contact area.<sup>46</sup> In the superior and anterosuperior quadrant, the labrum is less well adhered to the glenoid. In some incidences, the labrum has the appearance of a knee meniscus. The glenoid labrum in the anterosuperior quadrant can be quite variable, with sublabral foramen common. Absence of the anterosuperior labrum, in conjunction with a cordlike middle glenohumeral ligament, has been described as a Buford complex.<sup>47,48</sup> The labrum is the insertion of the glenohumeral ligaments and capsule. It is also the origin of the tendon of the long head of the biceps brachii muscle along the superior aspect.

The glenohumeral joint capsule has several areas of thickening that are called glenohumeral ligaments (Fig. 36.6). The inferior glenohumeral ligament is a hammocklike structure, with bands extending both anteriorly and posteriorly along the inferior aspect of the glenohumeral joint. The ligament originates along the inferior aspect of the humeral metaphysis and inserts onto the





**Fig. 36.6** A sagittal view of the glenohumeral joint showing the glenohumeral ligament, glenoid labrum, and biceps tendon. *Sup.*, Superior.

anteroinferior and posteroinferior glenoid labrum. The inferior glenohumeral ligament is considered one of the most important stabilizing structures in the abducted, externally rotated shoulder.<sup>49–51</sup> The glenoid insertion site of the anterior band of the inferior glenohumeral ligament consists of two attachments, one to the glenoid labrum and the other directly to the anterior neck of the glenoid.<sup>52</sup>

The middle glenohumeral ligament is the most variable of the glenohumeral ligaments and has been shown to be absent in up to 36% of persons.<sup>53</sup> It can vary from a thin fold of capsule to a thick cordlike structure. The middle glenohumeral ligament originates from the anterosuperior labrum or glenoid rim and crosses over the deep portion of the subscapularis prior to inserting onto the anatomic neck of the humerus. Although its presence and morphology vary, it has been shown to be important for anterior stability with the shoulder in 45 degrees of abduction.<sup>54</sup>

The superior glenohumeral ligament (SGHL) arises from the anterosuperior labrum, runs parallel to the biceps tendon in the rotator interval, and inserts onto the lesser tuberosity.<sup>55</sup> Fibers of the SGHL help create the biceps pulley, which stabilizes the biceps in the bicipital groove. The function of the SGHL has generated considerable debate, with one study reporting that the SGHL was the most important stabilizer for inferior translation.<sup>56</sup> Other investigators have concluded that the coracohumeral ligament is a more important stabilizer for inferior translation.<sup>57</sup> Yet other investigators have concluded that the inferior glenohumeral ligament is the most important stabilizer for inferior translation of the adducted shoulder.<sup>58</sup>

The rotator interval is a triangular space that is bordered inferiorly by the subscapularis tendon, anteriorly by the supraspinatus tendon, and medially by the glenoid. The rotator interval has been associated with shoulder instability, adhesive capsulitis, and isolated defects and tears. The rotator interval contains the coracohumeral ligament, which is a trapezoidal structure that originates on the lateral coracoid, traveling in two bands as it

inserts onto the lesser and greater tuberosities and over the bicipital groove. It also contains the SGHL, which arises from the superior labrum and supraglenoid tubercle, travels over the medial portion of the rotator interval, and blends with the coracohumeral ligament prior to inserting into the lesser tuberosity. The rotator interval also contains the joint capsule, which can be quite thin and variable. The rotator interval capsule can be as thin as 0.06 mm, and congenital defects are found in up to 75% of specimens.<sup>59</sup>

Sectioning of the coracohumeral ligament has been shown to result in increased inferior and posterior translation of the humerus, and imbrication and tightening of the coracohumeral ligament has been shown to decrease inferior and posterior translation.<sup>60</sup> During the past 15 years, as arthroscopic shoulder stabilizations have become more common, rotator interval closures have been advocated as an adjunctive procedure to help with postoperative stability, particularly when the instability is inferior and posterior. However, arthroscopic rotator interval closures have not been shown to be of benefit in adding posterior or inferior stability in a cadaveric model.<sup>61</sup>

### Glenohumeral Joint Capsule

The glenohumeral joint capsule is important in maintaining the intra-articular vacuum that helps to stabilize the joint. The joint capsule varies from 1.3 to 4.5 mm. It is thickest in the antero-inferior quadrant, which accounts for the anterior band of the glenohumeral ligament, and it is thinnest in the rotator interval and posterior quadrant.<sup>62,63</sup> A complex arrangement of the collagen fibers is present in the capsule, with a pattern of cross-linking in the superior capsule and a crossing pattern of spirals in the anterior and inferior capsule. The fibers in the ligamentous reinforcements radiate obliquely from the glenoid but vary greatly in orientation.<sup>64</sup> The inferior humeral attachment of the capsule may extend well below the articular surface. The inferior capsule has distinct internal and external folds.<sup>65</sup> Contracture of the glenohumeral joint capsule affects glenohumeral motion, which ultimately affect shoulder mechanics. This mechanism is commonly seen in patients with adhesive capsulitis. Another example is posterior capsular contractures, a diagnosis commonly seen in overhead throwers. Posterior capsular contracture results in a loss of internal rotation of the shoulder. The contracture and loss of internal rotation force the humeral head into a postero-superior position as opposed to the normal posteroinferior position found in a normal shoulder when externally rotating in the cocking phase.<sup>66</sup> Ruffini end organs and Pacinian corpuscles are found in the inferior, middle, and SGHLs.<sup>25</sup> It is possible that these mechanoreceptors signal the dynamic muscle stabilizers when the capsule is stretched in the abducted-external rotated position.

The tendon of the long head of the biceps brachii originates from the superior labrum and supraglenoid tubercle (see Fig. 36.6). Anatomic variations of the biceps origin have been described, including a bifid origin,<sup>67</sup> an extraarticular origin,<sup>68</sup> and origin from the rotator cable.<sup>69</sup> The biceps tendon has been shown to be an additional dynamic stabilizer of the shoulder. Loading the biceps tendon has been shown to decrease both anterior-posterior and superior-inferior translation<sup>70–72</sup> and also

has been shown to increase torsional rigidity against rotation force, limiting both external and internal rotation.<sup>72,73</sup> The biceps tendon travels out of the glenohumeral joint in the rotator interval prior to traveling distally in the bicipital groove of the proximal humerus. The tendon is stabilized in the proximal portion of the groove by the biceps pulley. The biceps pulley is a capsulo-ligamentous complex that is composed of fibers from the SGHL, the coracohumeral ligament, and the distal portions of the subscapularis tendon. More distally in the bicipital groove, the tendon is stabilized by the transverse humeral ligament. Recent evidence has suggested that the transverse humeral ligament is an extension of the subscapularis tendon (see Fig. 36.3).<sup>74</sup>

### Acromioclavicular Joint

The articulation between the distal end of the clavicle and acromion is called the acromioclavicular joint (see Fig. 36.3). The acromioclavicular joint is a synovial joint that consists of the articular facet of the acromion, the distal end of the clavicle, an intra-articular disk, and a joint capsule with thickening called the acromioclavicular ligaments. When viewed from the anterior position, the acromioclavicular joint is usually slightly medially or vertically oriented. With respect to the shaft of the clavicle, the inclination averages 12 degrees.<sup>75</sup> The joint is stabilized by the bony articulation but also by the acromioclavicular and coracoclavicular ligaments.<sup>76</sup> Several anatomic studies have assessed the location of the acromioclavicular ligaments and capsule.<sup>75,77,78</sup> Measurements in these studies are different, but one must be aware that the superior acromioclavicular ligament begins to insert as close as 2.3 mm from the lateral end of the clavicle.<sup>78</sup> The ligament does blend with the periosteum along the superior clavicle, and fibers of the ligament can be seen as far as 12 mm from the distal clavicle.<sup>77</sup>

It is generally accepted that superior-inferior stability of the acromioclavicular joint is due to the trapezoid and conoid ligaments, commonly referred to as the coracoclavicular ligaments (see Fig. 36.3). The lateral of the two, the trapezoid ligament, originates from the base of the coracoid, anterior and lateral to the conoid ligament. It has a broad insertion on the undersurface of the clavicle. The trapezoid ligament helps to reduce axial compression forces at the acromioclavicular joint.<sup>79</sup> The cone-shaped conoid ligament originates from the posteromedial aspect of the base of the coracoid and inserts on the conoid tubercle of the clavicle.<sup>80</sup> The conoid ligament is the most important ligament for superior-inferior stability of the acromioclavicular joint.<sup>79</sup> Anterior-posterior stability is also due to the coracoclavicular ligaments and the acromioclavicular ligaments. The trapezoidal ligament has been shown to provide the majority of restraint to the posterior translation of the clavicle.<sup>76,81</sup>

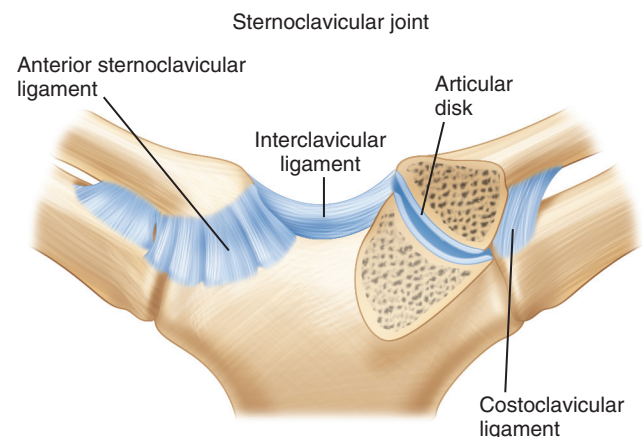
Some motion occurs at the acromioclavicular joint, but the motion is relatively small. Joint compression and translation occur as a result of protraction, retraction, and tilting of the acromion with overhead motion. Some clavicular rotation also occurs during abduction and adduction of the shoulder. Innervation of the acromioclavicular joint is from sensory branches from the suprascapular nerve.<sup>82</sup>

Distal clavicle resections are commonly performed, and this procedure can result in acromioclavicular joint instability.

Arthroscopic distal clavicle resection has been shown to increase acromioclavicular joint motion with a posterior applied force.<sup>83</sup> The acromioclavicular joint capsule is thickest superiorly and posteriorly. Preserving the superior and posterior acromioclavicular ligaments has been shown to be important in preserving joint stability when a distal clavicle resection is performed.<sup>84</sup> This preservation can usually be accomplished if less than 8 mm of bone is resected from the distal clavicle in men<sup>78</sup>; however, it is necessary to confirm that ligamentous tissue is present superiorly at the end of the procedure.

### Sternoclavicular Joint

The sternoclavicular joint is truly the only connection of the shoulder complex to the axial skeleton. The proximal end of the clavicle has an irregular surface, being concave in the antero-posterior plane and convex superiorly. The articular surface along the sternum is small and not congruent to the end of the clavicle. Like the acromioclavicular joint, the sternoclavicular joint also has an intra-articular disk. As a result of the joint incongruity, the articular surfaces offer no significant stability to this joint. The stability of the sternoclavicular joint is therefore due to its surrounding capsular and ligamentous support (Fig. 36.7). The capsular ligament covers the anterosuperior and posterior aspects of the joint and helps to stabilize the clavicle against abnormal translations. The posterior capsule is the main stabilizer of the sternoclavicular joint against anterior and posterior translations. The costoclavicular and interclavicular ligaments do not seem to have a significant anteroposterior stabilizing effect on the joint,<sup>85</sup> although the interclavicular ligament does help to stabilize the joint to superior translations. Most of the motion of the clavicle occurs at the sternoclavicular joint. Motion analysis has shown that the ligamentous support allows for up to 15 degrees of elevation, 30 degrees of retraction, and 30 degrees of rotation along the long axis of the clavicle with active arm elevation.<sup>86,87</sup> The medial end of the clavicle does have a physis, which is the last physis in the body to fuse. In one radiographic study, the earliest observation of complete fusion was at 26 years of age.<sup>88</sup>



**Fig. 36.7** A view of the sternoclavicular joint showing the anterior and posterior capsular ligaments, interclavicular ligament, articular disk, and costoclavicular ligament. The posterior capsular ligament provides the most stability against anterior and posterior translation of the sternoclavicular joint.

## Scapulothoracic Articulation

The scapular motion along the thoracic rib cage is referred to as the scapulothoracic articulation. Motion of the scapula is a result of the various stabilizing muscles that insert on it. The serratus anterior is an important stabilizing muscle because it holds the medial scapula to the thorax. It also helps to rotate and elevate the scapula in normal shoulder motions. The serratus anterior, along with the trapezius muscle, have the most muscle activity during arm abduction.<sup>89</sup> Smooth scapulothoracic motion is critical for normal shoulder kinematics. Alterations in scapular mechanics can lead to problems with glenohumeral instability and rotator cuff dysfunction. The biomechanics of normal shoulder motion, including scapular mechanics, will be discussed in more detail later in this chapter.

## SHOULDER MUSCLES

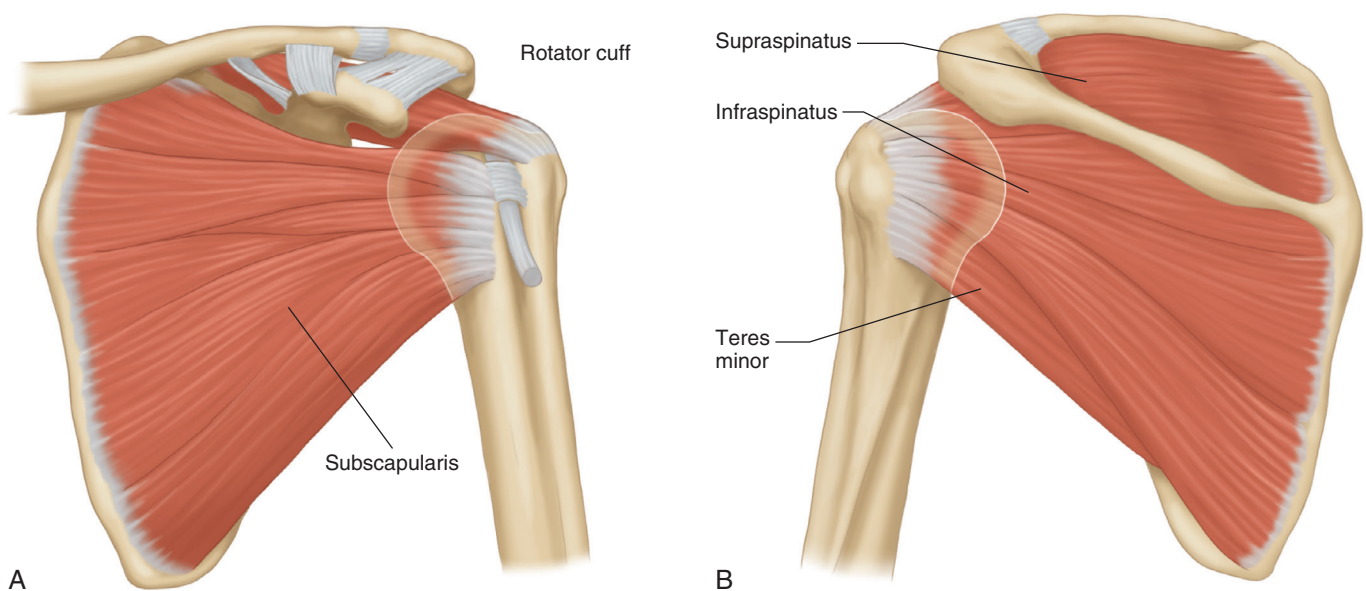
### Rotator Cuff

The rotator cuff consists of four muscles—the supraspinatus, infraspinatus, teres minor, and subscapularis (Fig. 36.8). The supraspinatus muscle originates in the supraspinatus fossa of the superior aspect of the scapula, passes in an anterolateral direction under the coracoacromial arch, and inserts on the greater tuberosity of the proximal humerus. The mean insertion footprint width of the supraspinatus is 14.7 mm, with the tendon insertion less than 1 mm lateral to the articular surface of the humeral head.<sup>90</sup> The supraspinatus muscle is innervated by the suprascapular nerve. The supraspinatus muscle is one of the main abductors of the humerus, accounting for 50% of the power to abduct in the scapular plane.<sup>91</sup> Other investigators have shown that paralysis of the supraspinatus and infraspinatus muscles results in a loss of 75% of abduction strength.<sup>92</sup> The anterior portion of the supraspinatus muscle contributes to

internal rotation in adduction and works as an external rotator as the shoulder is abducted.<sup>93</sup> Macroscopically the tendons of the rotator cuff interdigitate to form a common insertion on the humerus. The supraspinatus muscle, along with the infraspinatus and teres minor muscles, have an area of insertion on the greater tuberosity of 6.24 cm<sup>2</sup>.<sup>90</sup> The attachment site consists of a complex arrangement of collagen fibers capable of distributing tensile loads in various shoulder positions.<sup>94</sup> The cuff tendons have several layers, including a layer of tendon fibers that are both parallel and densely packed and another layer with obliquely oriented and loosely packed tendon fibers. The tendons also have layers composed of capsular and ligamentous tissue.<sup>95</sup>

The supraspinatus tendon has been shown to have a hypovascular area near the insertion,<sup>96–98</sup> which may play a factor in the pathogenesis of tears of the rotator cuff. Histologic sections from surgical specimens of patients with bursal-sided tears have been shown to be avascular, with no evidence of reparative tissue.<sup>99</sup> The articular side of the rotator cuff is subjected to and is more vulnerable to tensile forces, which might help to explain why partial rotator cuff tears occur more on the articular side compared with the bursal side.<sup>100</sup>

The main external rotators of the glenohumeral joint, the infraspinatus and teres minor muscles, originate on the posterior aspect of the scapula, with the infraspinatus muscle arising from the infraspinatus fossa (see Fig. 36.8). These muscles are inferior to the scapular spine and travel laterally to insert on the posterior aspect of the greater tuberosity. The infraspinatus muscle is innervated by the suprascapular nerve as it travels through the spinoglenoid notch. The teres minor muscle is innervated by the axillary nerve. The greater tuberosity has a small tubercle posteroinferiorly that is the site of the teres minor insertion. The infraspinatus muscle accounts for approximately 70% of the external rotation strength but also contributes approximately 45% of the abduction strength.<sup>92</sup> Clinically, the infraspinatus



**Fig. 36.8** Anterior (A) and posterior (B) views of the shoulder showing the subscapularis muscle and tendon anteriorly and the infraspinatus and teres minor muscles and tendons posteriorly. The supraspinatus is seen in the supraspinatus fossa. It travels anterolaterally to insert onto the greater tuberosity.



muscle is more functional at lower levels of abduction and elevation, whereas the teres minor muscle contributes more to external rotation power at higher levels of abduction and elevation.

The subscapularis is the anterior-most muscle of the rotator cuff; it originates from the ventral surface of the scapula and inserts on the lesser tuberosity. The lesser tuberosity insertion is trapezoidal in shape, with a broad insertion in the proximal portion. The upper 60% of the insertion site has a mean width of 2.5 cm.<sup>101</sup> As the subscapularis muscle travels laterally to insert, it passes under the coracoid process and the conjoined tendon of the short head of the biceps brachii and coracobrachialis. Narrowing of the distance from the lesser tuberosity to the coracoid may lead to subcoracoid impingement, which can lead to tearing of the subscapularis muscle.<sup>102</sup> The subscapularis muscle internally rotates and helps to adduct the humerus and is innervated by the upper and lower subscapular nerves. The subscapularis muscle has been shown to help provide stability to the shoulder from 0 to 45 degrees of abduction.<sup>54</sup>

The function of the rotator cuff, as its name implies, is to rotate the humerus. The rotator cuff also functions to maintain the center of rotation of the humeral head in the glenoid. The cuff muscles are important dynamic stabilizers of the glenohumeral joint. A 50% reduction in rotator cuff force has been shown to significantly increase anterior and posterior translation of shoulder specimens subjected to applied loads.<sup>103</sup>

Because of its orientation and vector of force, the rotator cuff also functions as a humeral head depressor. The head depressor effect is mainly due to the effect of the subscapularis and the external rotators. The supraspinatus muscle does not appear to have an effect on humeral head depression when it is studied biomechanically.<sup>104</sup> Although tears of the supraspinatus muscle can and do result in abnormal shoulder function, relatively normal kinematics can occur with active overhead motion with a balanced force couple of the subscapularis and infraspinatus/teres minor muscles.<sup>105</sup>

## Scapular Muscles

The scapular muscles are important to shoulder function because they ensure proper positioning of the scapula in conjunction with glenohumeral joint motion. The scapula, through its connections to the axial skeleton, is an important structure that is needed to help transfer energy from the lower extremities and lower torso to the upper extremities. The transfer of lower body energy to the shoulder is referred to as the “kinetic chain.”<sup>106</sup>

The scapular muscles stabilize the scapula, providing a stable base for shoulder movement. The scapular muscles include the serratus anterior, trapezius, levator scapulae, rhomboids, and pectoralis minor. Although all of these muscles have an important role, the serratus anterior and trapezius muscles appear to be most responsible for proper scapulohoracic motion.

The serratus anterior muscle originates from the upper thorax, along the ribs. It has a broad insertion along the anterior portion of the medial border of the scapula and is innervated by the long thoracic nerve (C5-C6-C7). The main function of the serratus anterior muscle is to protract the scapula laterally along the thorax with overhead motion of the shoulder. The serratus anterior muscle also works together with the upper and lower portions of

the trapezius muscle to rotate the scapula upward with overhead activities. The serratus anterior and upper and lower trapezius muscles have the most electromyographic-documented activity of the scapular muscles with arm elevation.<sup>89</sup>

The trapezius is a posterior muscle along the posterosuperior aspect of the scapula. It has three functional sections—the upper, middle, and lower trapezius. The superior trapezius originates from the occiput and inserts on the posterior portion of the lateral clavicle. The upper trapezius functions to help support the weight on the upper extremity. The middle portion originates from the spinous process of C7 to T3. The fibers of the middle portion insert onto the medial acromion and posteriorly along the medial portion of the scapular spine. The inferior trapezius originates from the spinous processes of T4 to T12 and travels superior and laterally to insert on the medial aspect of the scapular spine. The main function of the upper and lower trapezius is to upwardly rotate the scapula during overhead activity. The upper and lower trapezius also work with the middle trapezius to help retract the scapula. The trapezius muscle is innervated by the spinal accessory nerve, as well as the third and fourth cervical nerves.

The levator scapulae muscle originates from the transverse processes of the cervical vertebra (C1-C4). It inserts on the superior angle of the scapula and helps to rotate the interior angle of the scapula medially. The levator scapulae muscle works with the rhomboids and pectoralis minor to rotate the medial scapula downward. It is innervated by the third and fourth cervical nerves.

The rhomboid major and minor muscles originate from the spinous processes of the thoracic vertebra from T2 to T5 and insert on the medial border of the scapula. The rhomboid muscles are innervated by the dorsal scapula nerve, which arises from C5. The main function of the rhomboid muscles is to help stabilize the scapula along the thorax. The rhomboid muscles help to retract the scapula and, as such, are antagonists to the serratus anterior. The rhomboid muscles work with the levator scapulae muscle to elevate the medial scapula and rotate the scapula downward.

The pectoralis minor muscle originates from the anterosuperior aspects of the anterior portions of ribs three through five. The muscle travels laterally to insert on the medial aspect of the coracoid. The main function of the pectoralis minor muscle is to pull the scapula inferiorly and medially. It is innervated by the medial pectoral nerve.

## Other Muscles of the Shoulder Complex

The deltoid is the most superficial muscle covering the glenohumeral joint. It originates from the clavicle, acromion, and scapular spine and inserts on the deltoid tubercle on the lateral aspect of the midhumerus. The deltoid muscle provides a significant contribution to abduction of the humerus<sup>91</sup>; this function is provided by the anterior and middle portions of the muscle. The anterior portion of the deltoid muscle helps to flex the humerus, whereas the posterior portion of the deltoid muscle helps to adduct and extend the humerus. The deltoid muscle is innervated by the axillary nerve as it travels out of the quadrilateral space (the space bordered by the teres minor muscle superiorly, the teres major muscle inferiorly, the long head of the triceps brachii



medially, and the humeral shaft laterally). The posterior humeral circumflex artery also travels in this space. The axillary nerve travels around the lateral aspect of the shoulder as it innervates the deltoid muscle. The distance from the lateral acromion to the axillary nerve averages 5 cm but can be as little as 3 cm. The distance decreases with shoulder abduction.<sup>107</sup> Surgical dissections through the deltoid muscle can injure the nerve if the deltoid muscle is split too far distally.

The teres major muscle, along with the latissimus dorsi muscle, helps to extend, internally rotate, and adduct the humerus. The teres major muscle takes its origin from the inferior angle of the scapula and inserts on the medial aspect of bicipital groove. The latissimus dorsi muscle inserts on the medial aspect of the bicipital groove, just anterior to the teres minor muscle.

The latissimus dorsi muscle originates from the spinous processes of T7 to L5, the thoracolumbar fascia, the posterior iliac crest, and the inferior angle of the scapula. The muscle fibers travel superiorly, laterally, and anteriorly to insert on the medial aspect of the floor of the bicipital groove. In some persons, a conjoined tendon insertion with the teres major muscle occurs. The latissimus dorsi muscle helps to extend, adduct, and internally rotate the shoulder. It is innervated by the thoracodorsal nerve.

The pectoralis major muscle originates from the medial portion of the clavicle (clavicular head), the anterior surface of the sternum, the upper ribs, and the aponeurosis of the abdominal external oblique muscle (sternal head). The fibers from the clavicular head travel inferiorly and laterally. The sternal head fibers travel laterally. The two heads converge to form a common tendon that inserts onto the lateral aspect of the bicipital groove. The pectoralis major muscle is innervated by the lateral and medial pectoral nerves. The pectoralis major muscle helps to internally rotate, adduct, and flex the humerus.

The short head of the biceps brachii muscle and the coracobrachialis muscle originate from a common tendon at the tip of the coracoid. They are innervated by the musculocutaneous nerve, which pierces the muscles, on average, 5 cm distal to the coracoid. However, small branches of the nerve have been shown to innervate the muscle as close as 17 mm from the coracoid.<sup>108</sup> An appreciation of the neural anatomy and the variations is important to potentially avoid nerve injury during surgeries that retract the conjoined tendon or osteotomize the coracoid.

## BIOMECHANICS AND KINETICS OF NORMAL SHOULDER MOTION

Normal glenohumeral and scapulothoracic biomechanics are prerequisites for achieving optimal shoulder function. Normal shoulder motion consists of a coordinated pattern of scapulothoracic and glenohumeral motion. The acromioclavicular and sternoclavicular joints also play a role in normal shoulder motion by guiding and limited scapular motion. The scapula has to move in conjunction with the humerus to maintain the humeral center of rotation in the various glenohumeral positions. In the normal shoulder, the center of rotation of the humeral head varies little with elevation or flexion/extension in the horizontal plane.

Coordinated scapular motion requires the scapula to protract and retract with elevation and lowering of the arm. Historically, glenohumeral to scapulothoracic motion has been described as being a 2:1 ratio after the initial 30 degrees of elevation.<sup>109</sup> More accurate three-dimensional motion analyses have been performed to better define the normal scapulohumeral motion.<sup>86,87,110–116</sup> Recent evidence would suggest that the ratio of glenohumeral elevation to scapular upward rotation is 2.3:1 during elevation and 2.7:1 when lowering the arm from overhead. Scapula upward rotation and posterior tilting increase until maximum elevation.<sup>111</sup> Posterior tilting is needed to clear the acromion from the rotator cuff to avoid impingement with overhead activity. Motion during humeral elevation includes clavicle elevation, retraction, and posterior rotation; scapular upward rotation, internal rotation, and posterior tilting; and glenohumeral elevation with humeral external rotation.<sup>87</sup> The largest amount of scapular rotation occurs between 80 and 140 degrees of arm abduction.<sup>117</sup> Obligatory humeral external rotation occurs as the arm is elevated at or anterior to the scapular plane, partly because of the geometry of the humeral head and glenoid.<sup>87,110,118</sup> Internal rotation is required from maximum elevation posterior to the scapular plane.<sup>110</sup> Therefore one can easily see how capsular contractions that prevent rotation can adversely affect elevation and overall motion. Scapular dyskinesia is common in patients with shoulder instability and rotator cuff impingement syndrome.<sup>119–120</sup> In fact, alternations in shoulder biomechanics can result from abnormalities in the scapular muscles, alterations in cervical and thoracic posture, abnormalities in joint mobility (such as seen in arthrosis or instability), and fractures of the scapula or clavicle that affect the position of the scapula or the length of the clavicle.

## SUMMARY

The shoulder is a complex arrangement of four bones, four articulations, and several muscles, tendons, and ligaments. The shoulder complex requires healthy and ligamentously stable joints and numerous muscles working in synergy to allow the upper extremity to be placed in various positions and to perform activities that subject it to extremes of force and torque. The various articulations and muscles are kinetically linked, and abnormalities in any of the joints or muscles can result in pain, dysfunction, and possibly injury to other structures. An appreciation of the anatomy and biomechanics of the shoulder complex will be beneficial to clinicians in making diagnoses, applying safe and appropriate surgical procedures, and implementing appropriate rehabilitation protocols.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

## SELECTED READINGS

### Citation:

Boileau P, Walch G. The three-dimensional geometry of the proximal humerus. Implications for surgical treatment and prosthetic design. *J Bone Joint Surg Br.* 1997;79B(5):857–865.

### Level of Evidence:

V, basic science study

**Summary:**

In this cadaveric study, the authors showed that the articular surface of the proximal humerus varies with respect to the inclination, retroversion, size, and medial and posterior offset. Findings from this study have influenced design features of newer shoulder prostheses, which now allow surgeons to reproducibly replicate the native anatomy.

**Citation:**

Gerber C, Blumenthal S, Curt A. Effect of selective experimental suprascapular nerve block on abduction and external rotation strength of the shoulder. *J Shoulder Elbow Surg.* 2007;16(6):815–820.

**Level of Evidence:**

I, prognostic study

**Summary:**

In this well-designed and well-performed study, the authors showed that the infraspinatus muscle is responsible for 70% of the external rotation strength of the shoulder and that it also contributes to abduction strength. The authors also showed that the supraspinatus and infraspinatus muscles were responsible for 75% of the abduction strength in healthy volunteers.

**Citation:**

Matsumura N, Ikegami H, Nakamichi N. Effect of shortening deformity of the clavicle on scapular kinematics: a cadaveric study. *Am J Sports Med.* 2010;38(5):1000–1006.

**Level of Evidence:**

V, basic science study

**Summary:**

In this cadaveric study, the authors showed that shortening of the clavicle by 10% of the original length had adverse effects on shoulder kinematics.

**Citation:**

O'Brien S, Schwartz R, Warren R. Capsular restraints to anterior-posterior motion of the abducted shoulder: a biomechanical study. *J Shoulder Elbow Surg.* 1995;4(4):298–308.

**Level of Evidence:**

V, basic science study

**Summary:**

In this cadaveric study, the authors demonstrated that the primary restraint to anterior and posterior translation of the abducted shoulder is the inferior glenohumeral ligament complex.

**Citation:**

Turkel S, Panio M, Marshall J. Stabilizing mechanisms preventing anterior dislocation of the glenohumeral joint. *J Bone Joint Surg Am.* 1981;63A(8):1208–1217.

**Level of Evidence:**

V, basic science study

**Summary:**

In this cadaveric study, the authors showed that the subscapularis muscle provides anterior stability to the adducted shoulder, the middle glenohumeral ligament provides stability at 45 degrees of abduction, and the inferior glenohumeral ligament provides anterior stability as the shoulder approaches 90 degrees of abduction.

**Citation:**

Kibler WB. The role of the scapula in athletic shoulder function. *Am J Sports Med.* 1998;26(2):325–337.

**Level of Evidence:**

V, review article/expert opinion

**Summary:**

In this review article, the author thoroughly reviews the role of the scapula in normal shoulder biomechanics and in athletics involving overhead actions. The author introduces the concept of the kinetic chain (i.e., the transfer of lower extremity energy to the upper extremity) and reinforces the importance of the scapula in the kinetic chain.

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